Práctica 2: Sistema cardiovascular

Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica

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Información general



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Asignatura: **Modelado de Sistemas Fisiológicos**

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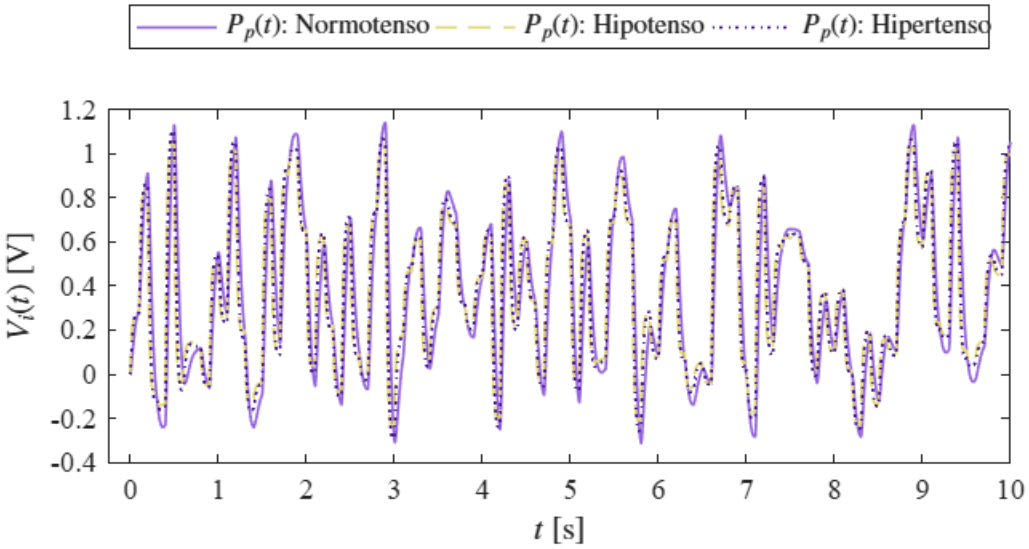
Datos de la simulación

```
clc; clear; close all; warning('off','all')
open_system('Sistema')
parameters.StopTime = '15';
parameters.Solver   = 'ode15s';
parameters.MaxStep  = '1e-2';
x = sim('Sistema',parameters);
t_data = (0:0.1:15)';
```

```
t = (0:0.001:15)';
Pa = interp1(t_data, x.Pa, t, 'previous');
writematrix(x.Pa,'signal.xlsx');
```

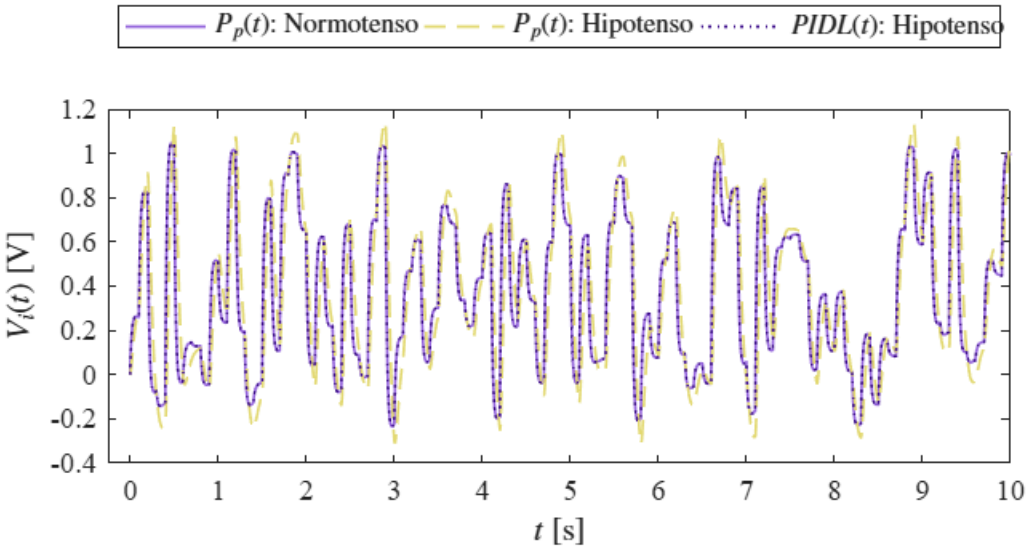
Respuestas: Hipotenso, Normotenso e Hipertenso

```
plotsignals(x.t, x.PpL, x.PpN, x.PpH, 'Individuos')
```



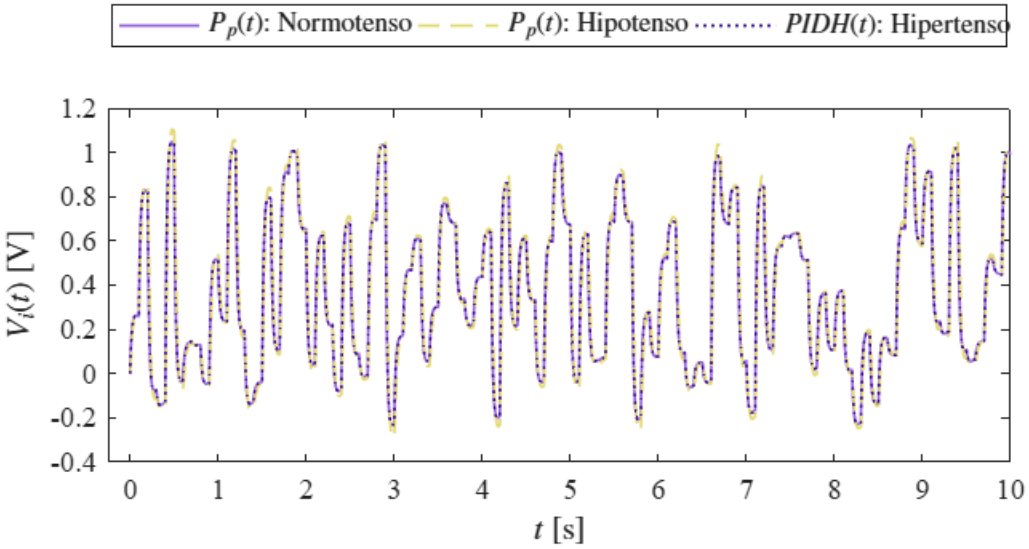
Respuestas: Normotenso, Hipotenso y Controlador

```
plotsignals(x.t,x.PpN,x.PpL,x.PIDL,'Hipotenso')
```



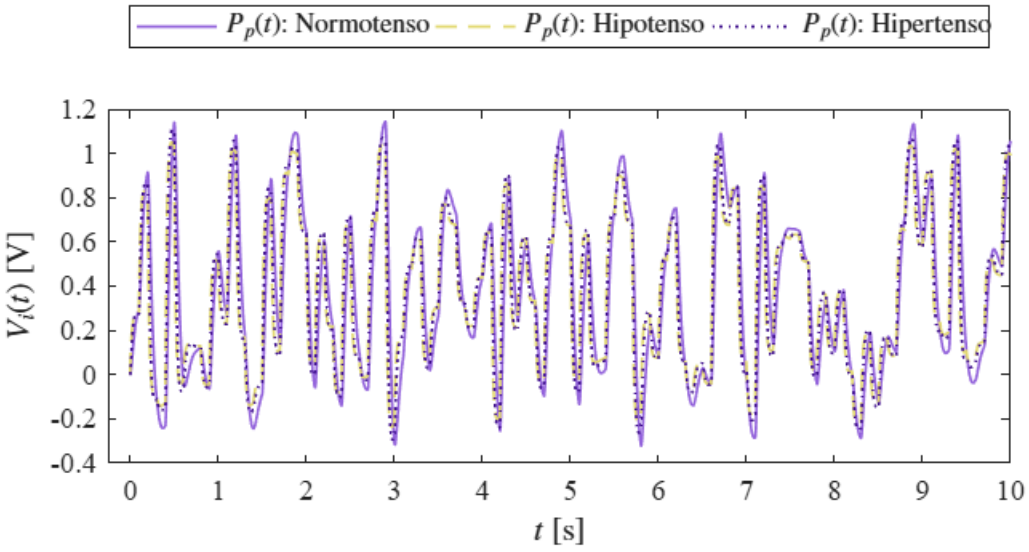
Respuestas: Normotenso, Hipertenso y Controlador

```
plotsignals(x.t,x.PpN,x.PpH,x.PIDH,'Hipertenso')
```



Modelo de EDOs

```
par_hipo = [0.01,0.001,1.00,3.500];
par_normo = [0.01,0.001,1.00,1.500];
par_hiper = [0.01,0.001,1.00,1.000];
PpN = cardio(par_normo,Pa,t);
PpL = cardio(par_hipo,PpN,t);
PpH = cardio(par_hiper,PpN,t);
plotsignals(t,PpL,PpN,PpH,'odes')
```




```
function Pp = cardio(par,Pa,t)
    Z = par(1); L = par (2); R = par(3); C = par(4); dt = t(2);
    n = round(t(end)/dt);
    FL = zeros(length(t),1); FL(1) = 0;
    Pp = zeros(length(t),1); Pp(1) = 0;
    for i =1:n
```

```
        FL(i+1) = FL(i) + (-Pp(i)/L + Pa(i)/L)*dt;
        Pp(i+1) = Pp(i) + (FL(i)/C - (1/(Z*C) + 1/(R*C))*Pp(i) + Pa(i)/(Z*C))*dt;
    end
end
```

```
function plotsignals (t,x1,x2,x3,signal)
set(figure(),'color','w')
set(gcf,'units','inches','position',[1,1,7,3.5])
set(gca,'fontname','times new roman','fontsize',11)
hold on; grid off; box on;
clr1 = ([149, 100, 221])/256;
clr2 = ([228, 218, 114])/256;
clr3 = ([62, 15, 141])/256;
order = [clr1;clr2;clr3]; colororder(order)
xlabel('$t$ [s]','Interpreter','latex')
ylabel('$V_i(t)$ [V]','Interpreter','latex')
xlim([-0.25 10]); xticks(0:1:10)

if signal == "Individuos"
    plot(t,x1,'-',t,x2,'--',t,x3,':','LineWidth',1)
    L = legend('$P_p(t)$: Normotenso','$P_p(t)$: Hipotenso','$P_p(t)$: Hipertenso');
elseif signal == "Hipotenso"
    plot(t,x1,'-',t,x2,'--',t,x3,':','LineWidth',1)
    L = legend('$P_p(t)$: Normotenso','$P_p(t)$: Hipotenso','$PIDL(t)$:
Hipotenso');
elseif signal == "Hipertenso"
    plot(t,x1,'-',t,x2,'--',t,x3,':','LineWidth',1)
    L = legend('$P_p(t)$: Normotenso','$P_p(t)$: Hipotenso','$PIDH(t)$:
Hipertenso');
elseif signal == "odes"
    plot(t,x1,'-',t,x2,'--',t,x3,':','LineWidth',1)
    L = legend('$P_p(t)$: Normotenso','$P_p(t)$: Hipotenso','$P_p(t)$: Hipertenso');


end

set(L,'interpreter','latex','location','northoutside','orientation','horizontal')
ylim([-0.4 1.2]); yticks(-0.4:0.2:1.2)

exportgraphics(gcf,[signal,'_matlab.pdf'],'ContentType','vector')

end
```